

The Same Rule In Water And In Air

SYNOPSIS

In Note 4a you found a rule with your ears and a tape measure: two in-phase sources make a place **loud** when the two path lengths differ by a whole number of wavelengths, and **quiet** when they differ by a whole number plus a half. One classroom is thin evidence for a claim that broad, and a classroom is a poor laboratory besides — you cannot switch off the walls, you cannot change the speed of sound, and you certainly cannot *see* the pattern. In this note you take the same rule into the PhET *Wave Interference* simulation, where the pattern is visible from above, where the frequency and the source separation are knobs you can turn, and where the same experiment runs in two completely different media: **transverse ripples on water** and **longitudinal sound in air**. If one rule governs both, you have learned something about waves rather than something about water.

BEFORE YOU OPEN THE SIMULATION — COMMIT TO AN ANSWER

Two taps drip into a water tank in step with each other, sending out circular ripples. You look down on the tank from above and watch for a while. **What does the pattern of calm water look like?**

- ☐ There is no calm water — the whole surface is choppy everywhere.
- ☐ A set of calm rings, centred on the midpoint between the taps.
- ☐ Calm lines that fan outward from between the taps, like spokes.
- ☐ A random, shifting mess with no fixed pattern.

Now say *why*, in one sentence, using the words “path” and “difference.”

Getting there

Search for “**wave interference phet**” and open *Wave Interference*. Choose the second panel, **Interference**. It opens on a water tank with two taps. Leave every control at its default for now: frequency in the middle, amplitude at maximum, separation 3 cm. Only the top tap is running.

TWO TOOLS YOU WILL USE CONSTANTLY

The **Water Level meter** (third item in the top right toolbox) has a **black** and a **grey** probe. Drop a probe anywhere in the tank and it traces the height of the water at that one point as time passes — exactly the “stand still and watch” view of Note 3, Fig. 2. The **measuring tape** gives you distances in centimetres. Bright rings are **crests**; dark rings are **troughs**.

Part 1 — One source: get the numbers

Before two sources can be interesting, you need the three numbers for one. Run only the top tap.

- 1 Click the green button on the top tap so the drops fall and ripples spread.
- 2 Check the **Graph** box and pause the run. Measure the distance from one crest to the next crest.

Wavelength, λ = _____ cm

- 3 Uncheck Graph. Put one probe in the tank and watch its trace. Find the **period** — the time from one peak of the trace to the next — then convert.

Period, T = _____ s

Frequency, $f = 1/T$ = _____ Hz

- 4 Read the **amplitude** from the trace. One vertical tick mark is 1 cm.

Amplitude, A = _____ cm

- 5 Compute the **speed** of the ripples. State the equation first.

v = _____ (equation)

v = _____ cm/s

Part 2 — One source, two probes: phase is distance

Still one tap. This part isolates the idea underneath everything else: how far a wave has travelled determines where it is in its cycle. Two points at different distances from the *same* source are at different phases, and you can see it directly on the

two traces.

- 1 Pause the run. Place the **black** probe on the first bright ring and the **grey** probe on the second bright ring. Restart. Are the two traces completely **in phase**?

In phase? _____

Why should you expect them to be? Answer before you move on.

- 2 Pause. Measure the distance from the drop point to each probe, and subtract.

to black: _____ cm **to grey:** _____ cm

$d_{\text{con1}} =$ _____ cm

- 3 Still paused, move the **grey** probe out to the **third** bright ring. Measure again and subtract.

$d_{\text{con2}} =$ _____ cm

- 4 How are d_{con1} and d_{con2} related to the wavelength you measured in Part 1? Write each one as a multiple of λ .

$d_{\text{con1}} =$ _____ λ $d_{\text{con2}} =$ _____ λ

- 5 Now find a position where the two traces are completely **out of phase**. Pause, put the black probe on the first bright ring and the grey probe on the first **dark** ring (not at the drop point). Restart and confirm.

Out of phase? _____

Why should you expect them to be?

- 6 Pause and measure both distances again.

$d_{\text{des1}} =$ _____ cm

- 7 Move the grey probe to the **second** dark ring and repeat.

$d_{\text{des2}} =$ _____ cm

8

Write d_{des1} and d_{des2} as multiples of λ .

$d_{\text{des1}} = \rule{1.5cm}{0.4pt} \lambda$ $d_{\text{des2}} = \rule{1.5cm}{0.4pt} \lambda$

9

Show, using your four numbers, that in-phase points satisfy $d = n\lambda$ with $n = 0, 1, 2, \dots$ — an integer number of wavelengths.

10

Show, using your four numbers, that out-of-phase points satisfy $d = (n + \frac{1}{2})\lambda$ — an integer plus a half.

WHAT YOU HAVE JUST DONE

Nothing in Part 2 involved two sources. You measured how the phase of *one* wave depends on how far it has travelled. That is the machinery. Two sources only make it visible, because they let you compare two different travel distances at the same instant, in the same place.

Part 3 — Two sources in water: the pattern appears

1

Turn on the **second** tap and let the waves from both sources fill the tank. Do not pause. Look at the whole tank from above and describe the pattern in one sentence. Then compare it with the prediction you committed to at the top of this note — were you right?

2

Place the **black** probe anywhere *except* a “fuzzy” band, and the **grey** probe in the middle of a fuzzy band. Watch both traces while it runs.

How can you tell from its trace that the black probe sits at **constructive** interference?

How can you tell that the grey probe sits at **destructive** interference?

3

Pause and measure four distances with the tape: from *each* tap to *each* probe. Fill in the table, then take the two differences.

	from top tap (cm)	from bottom tap (cm)	Δ = difference (cm)	Δ $\div \lambda$
black probe (loud/constructive)				
grey probe (calm/destructive)				

4

Show from your black-probe row that constructive interference occurs where $\Delta = n\lambda$. Then explain *why*, using the phase relationship between the two waves arriving there — not just the arithmetic.

5

Show from your grey-probe row that destructive interference occurs where $\Delta = (n + \frac{1}{2})\lambda$. Again, explain why in terms of phase.

Part 4 — The same experiment in air

Switch the simulation to **Sound**. The taps become speakers, the water surface becomes air, and the transverse ripples become longitudinal pressure waves. Almost nothing about the physical situation is the same. Leave all settings at their defaults.

1

With one speaker running, measure the **wavelength** of the sound wave.

$\lambda =$ _____ cm

2

Turn on both speakers. Put the black probe in a non-fuzzy region and the grey probe in a fuzzy one, and measure the four distances as before.

	from top speaker (cm)	from bottom speaker (cm)	Δ (cm)	$\Delta \div \lambda$
black probe (loud)				
grey probe (quiet)				

3

Do your two $\Delta \div \lambda$ values come out to a whole number and a whole-number-plus-a-half, the same way the water values did?

4

List three things that are *physically different* between the water case and the sound case — the medium, the type of wave, the speed, the wavelength, what is actually oscillating. Then state, in one sentence, what stayed the same.

$$\Delta = n\lambda \rightarrow \text{constructive} \quad \cdot \quad \Delta = (n + \frac{1}{2})\lambda \rightarrow \text{destructive}$$

Two sources in phase with each other, same frequency, same medium — and now verified in **two different media**. The rule does not care what is waving. It cares only about path difference measured in wavelengths.

Part 5 — Testing your Note 4a predictions

At the end of Note 4a you predicted what happens to the spacing of the quiet spots when the frequency changes and when the sources move closer together. The classroom could not test either one. This simulation can.

1

Frequency. Keeping the separation fixed, raise the frequency. What happens to λ ? What happens to the spacing of the calm bands? Record both.

λ : _____ **band spacing:** _____

Did your Note 4a prediction (Q11) hold?

2

Separation. Return the frequency to its default and change the **separation** of the two sources. Move them closer. What happens to the spacing of the calm bands?

band spacing: _____

Did your Note 4a prediction (Q12) hold?

3

Set the separation to its **smallest** value, so the two sources are nearly on top of each other. What happens to the pattern, and why? What is the largest path difference Δ that two nearly-coincident sources can possibly produce?

4

The centre line, running straight out between the two sources, was loud in the classroom and is bright in the simulation, at *every* frequency and *every* separation. Explain why that one line is immune to both knobs.

Performance Expectations

Measure wavelength, period, frequency and amplitude of a wave from a simulation, and compute the wave speed from them. Explain why two points at different distances from a single source are at different phases, and determine from measured distances whether two points are in phase or out of phase. Determine experimentally that constructive interference occurs where the path difference from two in-phase sources is an integer number of wavelengths and destructive interference where it is an integer plus a half, and justify each using the phase relationship of the arriving waves. Demonstrate that the same path-difference rule governs a transverse wave in water and a longitudinal wave in air, and identify what is physically different and what is conserved between the two cases. Predict and then verify how the interference pattern responds to a change in frequency and to a change in source separation.

WHERE THIS IS GOING

Interference is the first of the three behaviours that only waves have; diffraction and refraction are the other two, and both are coming. But notice what you have also quietly built: a *measuring instrument*. If you know the geometry and can find a quiet spot, you can work backwards to the wavelength — and from the wavelength, to the frequency or the wave speed. That trick is how the wavelength of light was first measured, long before anyone could see a light wave, and it is the whole basis of Lab 2.

Files for this note

Files for this note. The simulation is at phet.colorado.edu/en/simulations/wave-interference — choose the **Interference** panel. The read PDF and the printable handout live at

byungmooncho.github.io/PHY105/handouts/M1_N4b_Interference_You_Can_See.pdf

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